Listing Even Cycles Faster than the Submodular-Width Barrier

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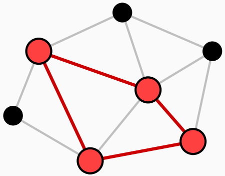
The Problem

Input
An undirected graph $G = (V, E)$, $m := |E|$, and a fixed $k \geq 2$.

Output
All copies of C_{2k} in G i.e. all subgraphs isomorphic to C_{2k}. We write $t := \#C_{2k}(G)$.

Goal: output-sensitive time

$$\widetilde{O}(f(m) + t), \qquad \text{minimise } f.$$



One copy of C_4 inside G

Why *Fixed* Length, and Why *Even*?

- **Fixed length.** Length n is Hamiltonian Cycle [Kar72], and hardness persists for $\ell = n^{\Omega(1)}$ [GJ79]. But the problem is FPT in ℓ [Mon85; AYZ95], so constant ℓ is tractable.

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This emergent structure is exactly what lets us beat generic algorithms.

Where Things Stood

Problem	Best known bound	Reference / comment
Detection of C_{2k} in terms of m	$O(m^{2k/(k+1)})$	Dahlgaard–Knudsen–Stöckel [DKS17]; conditionally optimal [LV20].
Listing C_4	$\widetilde{O}(\min\{n^2, m^{4/3}\} + t)$	Jin–Xu [JX23]; Abboud–Khoury–Leibowitz–Safier [Abb+23]; conditionally optimal.
Listing C_6	$\widetilde{O}(m^{8/5} + t)$	Vassilevska Williams–Westover [WW25]; computer-aided proof (576 cases).
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We improve those. **Re-proved in the general-k framework.**

Our Result

All $2k$-cycles of an undirected m-edge graph can be listed in time

$$\widetilde{O}\left(m^{\frac{2k^2-k+1}{k^2+1}}+t\right) \;=\; \widetilde{O}\left(m^{2-\frac{1}{k}-\frac{k-1}{k(k^2+1)}}+t\right).$$

- First improvement over AYZ for every $k \geq 3$ — open for 30 years.
- $k=3$: same exponent $8/5$ as [WW25], but no $\log^{74} n$ factor, no 576-case computer search, no BFS.
- Key ingredient: an extremal graph theory statement for even cycles.

Extremal Graph Theory

Unbalanced Supersaturation

- **Simonovits; Jiang-Yepremyan :** [JY20]

$$m \gtrsim n^{1+1/k} \quad \Longrightarrow \quad \#C_{2k}(G) \gtrsim \left(\frac{m}{n}\right)^{2k}$$

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- **Unbalanced Supersaturation Conjecture:** [Jin–Zhou, in [WW25]]

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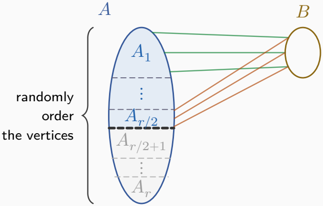
- **Our theorem:** $m \gtrsim a\, b^{1/k} \quad \Longrightarrow \quad \#C_{2k}(G) \gtrsim m^{2k}/a^k b^k$

Proof overview



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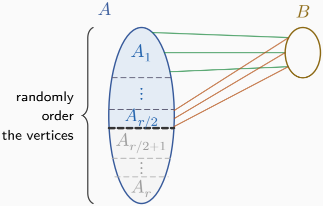

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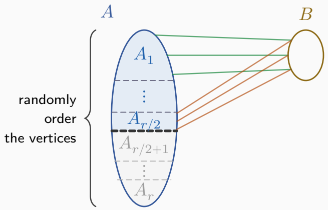

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- **Randomise.** A fixed partition misses cycles whose A-vertices split across buckets, so order A uniformly at random.

The Algorithm

Step 0: The Reduction

Global listing $\le_{\text{Cook}}$ Anchored listing

Global listing

Given G:
list *all* C_{2k}

$\xrightarrow[\text{subroutine}]{\text{calls as}}$

Anchored listing

Given G and v:
list all C_{2k} through v

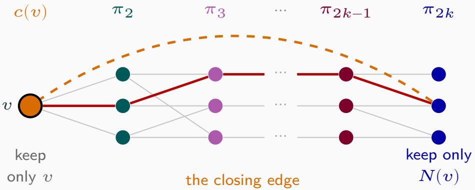
$\widetilde{O}(\,?\, + t)$

$\widetilde{O}(|E| + t_v)$ per call

Step 1: Algorithm for Anchored Listing

Input $(G, v) \longrightarrow$ **all** C_{2k} **through** v

1. **Colour coding** [AYZ95]. Random $2k$-colouring: a fixed C_{2k} is colourful w.p. $2^{-O(k)}$, so $\widetilde{O}(1)$ repetitions reduce to listing *colourful* cycles.
2. **Layered graph.** For each colour order π: layer $1 = \{v\}$, layer $j = \pi_j$, last layer $\cap\, N(v)$; edges only between consecutive layers.
3. **Enumerate.** DFS all $v \to$ last-layer paths; deduplicate in $O(1)$.

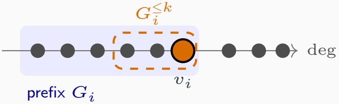


Running time: $\widetilde{O}(|E| + t_v)$**.**

Step 2: Algorithm for Listing All C_{2k}

Input $G \longrightarrow$ **all** C_{2k}

1. Order the vertices by *increasing degree*, $v_1 \prec v_2 \prec \cdots \prec v_n$.
2. Prefix graph $G_i \coloneqq G[\{v_1, \ldots, v_i\}]$.
3. $G_i^{\leq k}$: the vertices of G_i within distance k of v_i; write $E_i \coloneqq E(G_i^{\leq k})$.
4. Call **Algorithm 1** on $(G_i^{\leq k}, v_i)$, for every i.



Correctness. Every C_{2k}'s $\prec$-largest vertex owns it $\Rightarrow$ listed exactly once.

$$T \;=\; \sum_{i=1}^{n} \widetilde{O}(|E_i| + t_i) \;=\; \widetilde{O}\Big(\underbrace{\textstyle\sum_{i=1}^{n} |E_i|}_{\text{all that is left}} + t \Big).$$

Step 3: The Running Time

Bound $\sum_{i=1}^{n} |E_i|$

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degree split at Δ

Capped k-walks: $x_0, \dots, x_k$ with $\deg(x_0)$ maximal

$$\sum_i |E_i| \quad \lesssim \quad \frac{m^2}{\Delta} + W, \qquad W \;:=\; \#\{\text{capped } k\text{-walks}\}$$

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Φ-norm [DKS17] + dyadic decomposition

An edge bound between vertex sets

$$|E(A,B)| \lesssim |A||B|^{1/k} + |B||A|^{1/k} + t^{\frac{1}{2k}} \sqrt{|A||B|}$$

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the extreme input

Weak unbalanced supersaturation

Unbalanced supersaturation conjecture

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$$T(\Delta) \lesssim \widetilde{O}\left(\frac{m^2}{\Delta} + m^{\frac{3k-1}{2k}} \Delta^{\frac{(k-1)^2}{2k}} + t\right)$$

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Back to Databases

Submodular Width

- Listing C_{2k} is a full conjunctive query:
  $C_{2k}(v_1, \ldots, v_{2k}) \leftarrow \bigwedge_i E(v_i, v_{i+1}) \wedge \bigwedge_{i \neq j} v_i \neq v_j$
- CQs are modeled as hypergraphs. Marx introduced the submodular width $\mathrm{subw}(Q)$ [Mar13]; PANDA [ANS25] evaluates any CQ in time $\tilde{O}\big(N^{\mathrm{subw}(Q)} + t\big)$
- For all subquadratic patterns, this is optimal in the (harder) coloured variant [BG25]
- Colour coding reduces C_{2k} to the closed-walk query W_{2k} (drop $\neq$), and $\mathrm{subw}(W_{2k}) = 2 - 1/k$ recovers the AYZ exponent

Why We Are Allowed to Beat It

- Beating AYZ = beating the submodular width for the cycle query
- Our exponent:

$$2 - \frac{1}{k} - \frac{k-1}{k(k^2+1)} \; < \; 2 - \frac{1}{k} = \mathrm{subw}(W_{2k})$$

- Possible because our query has self-joins + symmetric input relation $E \Rightarrow$ kills the adversarial gadgets behind the [BG25] lower bound

Open Problems

- The full Unbalanced Supersaturation Conjecture. It would give the optimal $\widetilde{O}(m^{2k/(k+1)} + t)$.
- The exponent is not tight. At $k = 2$ our formula gives $m^{7/5}$, while the truth is $\Theta(m^{4/3})$.
- More general conjunctive queries. Which classes with self-joins / symmetric relations beat submodular width?
- Supersaturation beyond graphs. Analogues for hypergraph patterns and higher-arity relations.

The End

Thank you ☺

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